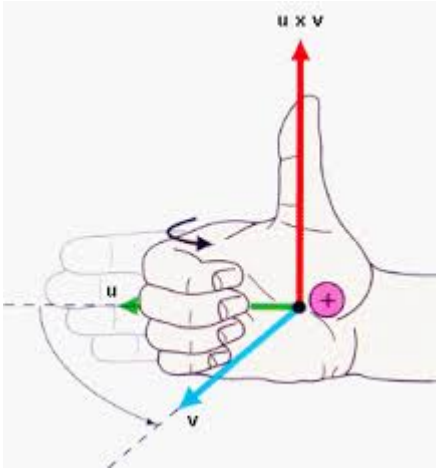
11.4 The Cross Product

- The *cross product* of two vectors is used to describe how a plane is tilting in space.
- If **u** and **v** are non-parallel vectors in space, then they determine a plane. So we can define **u**×**v** to be the vector perpendicular to this plane. (See the definition later.)
- We will use the *right-hand rule*.
- Also called the *vector product*.



Definition **Cross Product**

$$\mathbf{u}\times\mathbf{v}=(|\mathbf{u}||\mathbf{v}|\sin\theta)\mathbf{n}$$

θ =the angle between **u** and **v**
n = the unit vector perpendicular to the plane determined by **u** and **v** according to the right-hand rule

If $\mathbf{u}=\mathbf{0}$ or $\mathbf{v}=\mathbf{0}$, then $\mathbf{u}\times\mathbf{v}=\mathbf{0}$.

If **u** is parallel to **v**, we also have $\mathbf{u}\times\mathbf{v}=\mathbf{0}$.

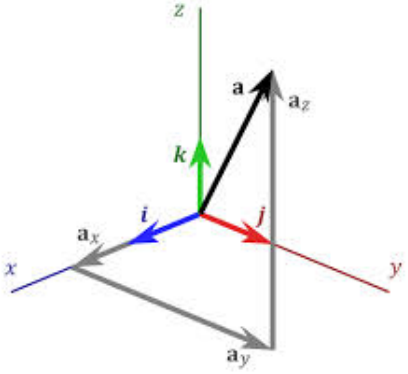
Properties of the Cross Product

Let **u**, **v**, **w** be vectors and let $r,\ s$ be scalars. Then,

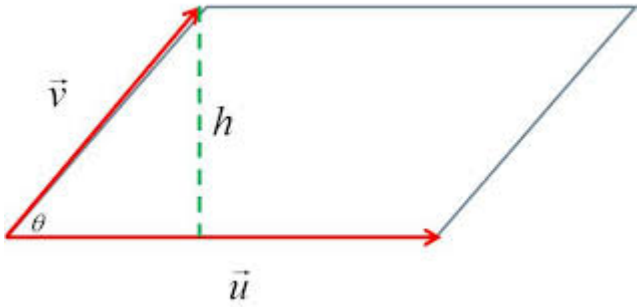
1. $(r\mathbf{u})\times\ (s\mathbf{v})\ =\ (rs)(\mathbf{u}\times\mathbf{v})$
2. $\mathbf{u}\times\ (\mathbf{v}+\mathbf{w})\ =\mathbf{u}\times\mathbf{v}+\mathbf{u}\times\mathbf{w}$
3. $(\mathbf{u}+\mathbf{v})\ \times\mathbf{w}=\mathbf{u}\times\mathbf{w}+\mathbf{v}\times\mathbf{w}$
4. $\mathbf{v}\times\mathbf{u}=\ -\ (\mathbf{u}\times\mathbf{v})$
5. $\mathbf{0}\times\mathbf{u}=\mathbf{0}$
6. $\mathbf{u}\times(\mathbf{v}\times\mathbf{w})=\ (\mathbf{u}\cdot\mathbf{w})\mathbf{v}-\ (\mathbf{u}\cdot\mathbf{v})\mathbf{w}$

Remark:

1. In general, $(\mathbf{u}\times\mathbf{v})\times\mathbf{w}\neq\mathbf{u}\times(\mathbf{v}\times\mathbf{w})$. Cross products are not associative.
2. Writing $\mathbf{u}\times\mathbf{v}\times\mathbf{w}$ is meaningless, a pair of parentheses should be included.


z
a
$\mathbf{a}_z$
k
i
j
$\mathbf{a}_x$
x
y
$\mathbf{a}_y$

|u×v| is the Area of a Parallelogram



Determinant Formula of u×v

$$\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$$

$$\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$$

u×v =

Example: Find the cross products $\mathbf{u}\times\mathbf{v}$ and $\mathbf{v}\times\mathbf{u}$ for

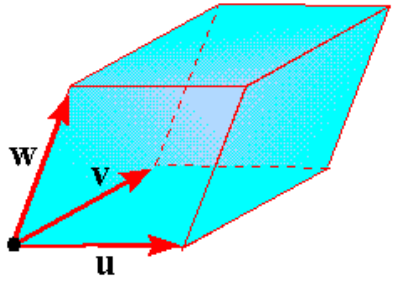
$$\mathbf{u} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$$

$$\mathbf{v} = -4\mathbf{i} + 3\mathbf{j} + \mathbf{k}$$

Example: Find a unit vector perpendicular to the plane containing $P(1, -1, 0)$, $Q(2, 1, -1)$, $R(-1, 1, 2)$. Also, find the area of the triangle ΔPQR.

Triple Scalar or Box Product

The product $(\mathbf{u}\times\mathbf{v})\cdot\mathbf{w}$ is called the triple scalar product of **u**, **v**, **w**.



Triple Scalar Product as a Determinant

$(\mathbf{u}\times\mathbf{v}) \bullet \mathbf{w} =$

Example: Calculate the volume of the box (parallelepiped) determined by vectors

$\mathbf{u} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$

$\mathbf{v} = -2\mathbf{i} + 3\mathbf{k}$

$\mathbf{w} = 7\mathbf{j} - 4\mathbf{k}$

Remark:

- $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ are coplanar $\Leftrightarrow (\mathbf{u}\times\mathbf{v})\bullet\mathbf{w} = 0$.

- If any of the two vectors $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ are the same, then the triple scalar product is zero. That is,

$$(\mathbf{u}\times\mathbf{u})\bullet\mathbf{v} = 0,$$
$$(\mathbf{u}\times\mathbf{v})\bullet\mathbf{u} = 0,$$
$$(\mathbf{u}\times\mathbf{v})\bullet\mathbf{v} = 0.$$

Example:

Find the volume of the parallelepiped having

$\mathbf{u}=\mathbf{i}+2\mathbf{j}-\mathbf{k}$, $\mathbf{v}=-2\mathbf{j}+3\mathbf{k}$ and $\mathbf{w}=\mathbf{i}+7\mathbf{j}-4\mathbf{k}$ as its sides.

Example:

Show that the points $P(0, 0, 3)$, $Q(2, -1, 2)$, $R(3, 2, 1)$ and $S(1, 3, 2)$ are coplanar.

Example:

Let $P(1, 2, 3)$, $Q(2, 4, 2)$, $R(1, 3, 2)$ and $S(a, 3, b)$ be four points in space and let $\vec{RS}$ be a unit vector. Find all possible coordinates of S if the volume of the parallelepiped determined by the vectors $\vec{PQ}$, $\vec{PR}$ and $\vec{PS}$ is 1 unit.

Example:

Show that for any vectors $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$,

$\mathbf{u}\times(\mathbf{v}\times\mathbf{w})+\mathbf{v}\times(\mathbf{w}\times\mathbf{u})+\mathbf{w}\times(\mathbf{u}\times\mathbf{v})=\mathbf{0}$.